The Validation Bottleneck: Alpha Discovery When Hypotheses Are Free

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Abstract

Large language models have made investment hypotheses effectively free. This paper argues that the binding constraint on alpha discovery has therefore moved from generation to validation, and that both the statistics and the economics of quantitative research reorganize around the one input that cannot be manufactured: out-of-sample time the generating model has provably not seen. We formalize a contamination result—temporal segregation of an LLM's data feed does not segregate its knowledge, so the model's training cutoff is the only valid out-of-sample boundary—and re-found an anti-HARKing research protocol on the theory of adaptive data analysis, in which mechanism-implied auxiliary predictions purchase statistical power from the cross-section of implications rather than from calendar time. On the economic side, the classical extreme-value hurdle implies an equilibrium alpha floor that rises logarithmically in the industry-wide count of hypotheses tested, a knowability constraint under which rapidly decaying alpha cannot be certified within its own lifetime, and a generator-monoculture channel through which shared foundation models correlate discovery itself across firms. Two pre-registered experiments on public data supply the evidence. When frontier models are asked to roleplay past dates and propose "novel" predictors, their proposals match the post-date anomaly literature at up to 3.3 times the base rate, and proposal frequency is *negatively* related to post-date factor performance (combined $z = -3.13$): the models reproduce the literature's celebrated, already-decayed factors rather than anticipating returns. When models with verified training cutoffs generate composite factors in an enumerable grammar—making the number of trials exact rather than estimated—and are evaluated strictly after their cutoffs, nothing survives dependence-robust false-discovery control: the best machine-generated strategy attains $t = 2.69$, below the expected maximum of an equal number of pure-noise strategies, and mechanism-implied predictions hold at coin-flip rates. Hypotheses are free, and worth what they cost until validated.

Keywords: large language models, backtest overfitting, data contamination, multiple testing, market efficiency, factor investing.

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Contents

1 Introduction

For as long as quantitative investing has existed, its implicit production function has treated ideas as the scarce input. Hypotheses were costly—they required researchers, reading, and time—while the data on which hypotheses were tested was a common resource, drawn on freely by every participant. The institutions of the field reflect this arrangement. Journals referee the plausibility of mechanisms; firms hire researchers by their ability to originate signals; the multiple-testing literature polices the boundary where cheap variations on ideas meet a finite data supply [Harvey et al., 2016; Bailey and López de Prado, 2014; Harvey and Liu, 2021].

Large language models invert the arrangement. A frontier model, prompted appropriately, returns eight coherent factor hypotheses—each with a named construction, an economic mechanism, and testable implications—in under a minute, at a marginal cost measured in cents. The generation of investment hypotheses, as an economic activity, has collapsed to a price of approximately zero. This paper is about what that collapse does to the statistics and the economics of alpha discovery. Our thesis is that the binding constraint moves to the one input that no model, vendor, or budget can manufacture: *out-of-sample time*—observations that postdate the knowledge of whatever process generated the hypothesis. When hypotheses are free, validation capacity is the scarce factor, and the field reorganizes around it. We call this the *validation bottleneck*.

The argument proceeds in four steps. The first step is epistemic (Section 2). It is tempting to evaluate machine-generated strategies the way human-generated strategies have always been evaluated: on historical data. For an LLM this is invalid in principle, not merely biased in practice. An LLM trained through 2024 has read the anomaly literature in its entirety—every published predictor, every replication, every post-publication decay estimate. Restricting the model's *data feed* to 1963–1999 does not restrict its *knowledge* of what was discovered, published, and arbitraged away after 1999. Temporal segregation of data is not temporal segregation of knowledge. We state this as a contamination result and draw its operational corollary: the model's training cutoff is the only out-of-sample boundary that exists, and any evaluation period that predates the cutoff carries no evidential weight for the model's proposals. Moreover, by McLean and Pontiff [2016], the literature the model has absorbed documents predictability that substantially decayed upon publication. The model has, in a precise sense, read only the obituaries: its prior over "what works" is a prior over what *worked*, weighted by prominence rather than by surviving profitability. This observation yields a falsifiable prediction that our first experiment confirms.

The second step is statistical (Section 3). If genuine out-of-sample time begins only at the training cutoff, then for current frontier models the valid evaluation window is a few years at most—and calendar time purchases statistical power at the punishing rate $t \approx \mathrm{SR}\sqrt{T}$: a strategy with a true annualized Sharpe ratio of 0.5 requires roughly 36 years of forward data to clear $t > 3$. Waiting is not a research design. We develop two responses. First, the validation process is an instance of *adaptive data analysis*: each time a generating process learns from validation outcomes, the holdout degrades, and the reusable-holdout theory of Dwork et al. [2015] supplies a principled budget of adaptive queries per validation window—machinery that, to our knowledge, has not previously been applied to financial research. Second, and centrally, a hypothesis that carries a genuine economic mechanism implies more than a mean return: it implies regime interactions, cross-sectional gradients, correlation structure, and replication patterns, each testable *simultaneously* on the same short window. Under independence, k auxiliary implications tested at sizes $\alpha_1, \dots, \alpha_k$ leave a false mechanism a survival probability of at most $\prod_i \alpha_i$. Power accumulates in the cross-section of implications rather than the time series of returns. This is, we argue, the only affordable source of statistical power in the bottleneck, and it explains why an anti-HARKing protocol for machine-scale research must require auxiliary predictions rather than merely permit them. A companion subsection establishes the underlying statistical geography: selection inflation is breadth-independent—the expected maximum backtest Sharpe among M null strategies is $\sqrt{2\ln M/T}$ whether one times a single index or ranks three thousand stocks—so protection against overfitting comes not from breadth per se but from capacity ratios, internal replication, and the effective length of the sample, each of which the protocol engineers deliberately.

The third step is economic (Section 4). In the equilibrium of Grossman and Stiglitz [1980], prices leave behind exactly enough predictability to compensate costly information acquisition. When acquisition—hypothesis generation—becomes free, the compensated activity shifts. We argue that equilibrium predictability comes to compensate *validation capacity*, and that the classical extreme-value threshold converts directly into an equilibrium object: with M hypotheses tested industry-wide, the t-statistic hurdle grows as $\sqrt{2\ln M}$, so the minimum alpha that can be distinguished from the best of M flukes—and hence the alpha at which capital allocation settles—rises logarithmically in M relative to effective validation time. The underlying inequality is not ours: it is the minimum-backtest-length result of Bailey et al. [2014] and appears as the exploitability threshold in Da et al. [2024], who also supply an equilibrium-compensation reading of statistical limits. Our contribution is the economic reinterpretation—M as the endogenous, machine-inflated industry trial count, T as scarce forward time rather than a historical estimation sample, and the resulting floor as the settling point of the funding process. Two further results follow. A *knowability constraint*: if alpha decays with half-life h while certification requires a threshold t^*, then under exponential decay the maximum attainable t-statistic is approximately $0.77\,\mathrm{SR}_0\sqrt{h}$, so strategies with $h \lesssim 1.7\,(t^*/\mathrm{SR}_0)^2$ years can never be certified within their own lifetime—validation structurally selects slow-decaying alpha, and the fastest-decaying opportunities are unknowable in real time. And a *generator monoculture* channel: when many firms draw hypotheses from a handful of shared foundation models, the correlation that the financial-stability literature locates in trading behavior [Kleinberg and Raghavan, 2021; European Central Bank, 2024; Dou et al., 2025; Meng and Chen, 2026b] moves upstream into discovery itself, collapsing the industry's effective number of independent trials while inflating its nominal one.

The fourth step is empirical (Section 5). Both experiments were pre-registered by design: generation corpora, evaluation walls, and analysis specifications were committed, in that order, before any performance data was examined, and every raw model response is archived. Experiment 1 measures the contamination directly. Three frontier models were prompted to roleplay five past dates (January 1990 through January 2010) and propose predictors "not yet documented in the academic literature" as of each date; the resulting 1,650 proposals were matched against the 212-predictor open-source anomaly library of Chen and Zimmermann [2022]. Proposals match predictors *published after the roleplay date* at up to 3.3 times the library base rate, with the enrichment rising as the vintage advances. The sharper test asks whether proposal frequency loads on post-date factor *returns*—information no truthful roleplayer could possess. It does not: the relation is negative (rank correlation negative in 13 of 15 model-vintage cells; Fisher-combined $z = -3.13$), while citation counts attract proposals. The models do not anticipate returns; they reproduce prominence. Experiment 2 is the constructive counterpart: the only design the contamination result permits, executed. Models with *verified* training cutoffs—we measure each model's effective cutoff with a recall-based probe and find knowledge extending up to five months past the reported dates—generated 1,008 composite factors inside an enumerable grammar over the same public library, making the number of distinct trials exact ($M = 330$ for the longest-window model). Evaluated strictly after the verified cutoffs, nothing clears dependence-robust false-discovery control at $q = 0.05$; the best strategy attains $t = 2.69$ against an extreme-value hurdle of 3.41—below the expected maximum of 330 strategies of pure noise—and mechanism-implied auxiliary predictions hold at 46%, the rate of a coin flip. On the one window where evidence is possible, machine-scale generation produced hypotheses worth exactly their production cost.

Section 6 draws the institutional conclusion. The data commons on which all of this testing operates is depletable, and adaptive machine-scale querying depletes it at industry scale; we describe a cross-firm testing registry whose unit of account is the reusable-holdout query budget, upgrading the trial-disclosure proposals of López de Prado [2019a] and the registered-reports proposal of Harvey [2017] from norms of disclosure into a mechanism of governance.

Related literature. Four strands intersect here; we take them in turn and state precisely what is inherited. The contamination strand documents lookahead bias in LLM-based prediction: Sarkar and Vafa [2024] define the problem and demonstrate it with unpredictable-event tests; Glasserman and Lin [2024] decompose in-window bias in sentiment backtests; Lopez-Lira et al. [2025] show that in-sample, "any observed output is consistent with both genuine skill and memorization"; Gao et al. [2025b] supply a portable in-sample diagnostic; Li et al. [2025] document agent backtest returns evaporating past the knowledge window. This literature concerns the evaluation of *given* forecasts; none of it treats hypothesis *generation*, where contamination enters through the literature the model has read rather than through recall of particular outcomes, and none develops the economics of the resulting scarcity—the closest statement is a single sentence in Gao et al. [2025b] noting that post-cutoff evaluation "limits statistical power." A mitigation strand seeks to engineer the bias away, by subtracting recallable outcome memory at inference time [Li et al., 2026], by approximate unlearning over an enumerable forget-set [Merchant and Levy, 2025], or by retraining chronologically clean models from scratch [He et al., 2025; Drinkall et al., 2024; Yan et al., 2026]. We engage these directly in Section 2: the first two operate on outcome-indexed memory and cannot reach selection on absorbed literature—Li et al. [2026] concede there is "no clean decomposition between memorised and reasoned components"—while clean retraining currently trades away the very capability that makes generation valuable. The multiple-testing and overfitting strand supplies our statistical raw material [Harvey et al., 2016; Bailey and López de Prado, 2014; Bailey et al., 2014, 2017; López de Prado and Bailey, 2018; López de Prado, 2019a]; the $\sqrt{2 \ln M}$ hurdle and the $2 \ln M/T$ length requirement are theirs, and our use is a reinterpretation, not a rederivation. The equilibrium strand is nearest in spirit: Martin and Nagel [2022] elevate out-of-sample tests epistemically in a high-dimensional learning economy; Dugast and Foucault [2025] price costly *search* for predictors in a Grossman–Stiglitz equilibrium—precisely the margin that generation-at-zero-cost eliminates; Da et al. [2024] derive equilibrium bounds from statistical learning limits on a fixed historical panel. None makes forward validation time the scarce input, none models the generation stage, and none treats the knowability constraint; those are the margins claimed here. Finally, Novy-Marx and Velikov [2026] industrialize the HARKing critique—LLMs writing persuasive theory around data-mined signals—and Si et al. [2025b,a] document mode collapse and the ideation–execution gap in LLM research ideas; both are complements: they establish that hypotheses are free and homogeneous, we establish what that makes scarce.

2 The Contamination Result and the Cutoff Wall

2.1 Temporal data segregation is not knowledge segregation

Consider a researcher evaluating a hypothesis-generating process G—human or machine—by the familiar protocol: form hypotheses, construct the implied strategies, and measure their performance on a historical sample $\mathcal{D}_{[t_0,t_1]}$. The protocol's validity rests on an exclusion restriction: the generation of the hypothesis must not have used information from the evaluation period. For human researchers this restriction is policed, imperfectly, by careers and norms; the multiple-testing literature exists because it fails at the level of the profession [Harvey et al., 2016]. For an LLM the restriction fails at the level of the individual hypothesis, by construction.

Let $\mathcal{I}_G$ denote the information set of the generator and let τ_G denote the end of its training corpus—the training cutoff. For a pretrained language model, $\mathcal{I}_G$ includes, in compressed form, the contents of its corpus: the empirical finance literature through τ_G, the financial press through τ_G, and extensive summaries of realized market history through τ_G. Two properties of $\mathcal{I}_G$ drive everything that follows.

First, $\mathcal{I}_G$ cannot be truncated by prompt. Instructing the model to "use only information available in 1999," or feeding it only pre-1999 data, restricts its *inputs*, not its *parameters*. The knowledge that value crashed in the late 2010s, that accruals decayed after Sloan [1996] was absorbed by practitioners, that low-risk anomalies were arbitraged as they became famous—all of this shapes which hypotheses the model finds plausible, whether or not any post-1999 number appears in its context window. The general-ML literature has established this directly: Gao et al. [2025a] show that prompting a model to adopt an earlier knowledge cutoff fails precisely when the excluded content is causally related to the query rather than directly asked for; Lopez-Lira et al. [2025] show that masking fails because models reconstruct entities and dates from minimal context. Hypothesis selection is the extreme case of causally related knowledge: the model is not asked to recall any post-cutoff fact, yet every candidate it ranks is ranked *because of* the absorbed record of what followed.

Second, $\mathcal{I}_G$ is not enumerable from outside. One can audit a human research program's trials, at least in principle [López de Prado, 2019a]; one cannot enumerate the strategy evaluations implicit in a trillion-token corpus. The effective number of hypotheses "already tested" inside the generator is unobservable, which is fatal for any multiple-testing correction applied to a pre-cutoff backtest of its proposals.

These observations combine into a statement we will use as a theorem-in-practice.

Proposition 2.1 (Contamination)**.** *Let G be a hypothesis generator with information set $\mathcal{I}_G$ containing the realized history of the evaluation sample $\mathcal{D}_{[t_0,t_1]}$, in any form—including descriptions, summaries, or analyses of that history. Then for hypotheses produced by G, performance statistics computed on $\mathcal{D}_{[t_0,t_1]}$ do not identify out-of-sample predictive content: the observed performance is consistent with any mixture of predictive skill and selection on $\mathcal{I}_G$, and the mixture is not identified because the selection channel is not observable.*

The logic is that of Lopez-Lira et al. [2025], transported from forecasts to hypotheses: in-sample, output consistent with skill is equally consistent with memory, and no statistic computed on the contaminated sample separates them. What makes the hypothesis case strictly harder is that the contamination does not reside in recallable (entity, date, outcome) triples that might be elicited and subtracted; it resides in the model's *ranking* of candidate mechanisms, i.e., in what mitigation approaches would classify as the reasoning to be preserved. The selection operator and the reasoning operator act on the same parameters; there is no decomposition to exploit.

Corollary 2.2 (The cutoff wall)**.** *The only sample on which performance statistics for G's hypotheses are interpretable is $\mathcal{D}_{(\tau_G^{\mathrm{eff}},\cdot]}$, where τ_G^{eff} is the generator's* effective *training cutoff. Genuine out-of-sample evidence begins at the wall and accumulates at calendar speed thereafter.*

Two features of the corollary deserve emphasis. First, it is an impossibility claim about the past, not a difficulty claim: no amount of pre-cutoff data, however long or however carefully segregated, contributes evidence. The 60-year panel that anchors the empirical anomalies literature is, for a 2024-trained generator, one long in-sample period. Second, the wall is a property of the *generator*, not of the strategy: the identical momentum–quality composite is testable on 1990s data if proposed by a process that has never read about the 1990s, and untestable there if proposed by a frontier LLM. Validity attaches to the pair (hypothesis, provenance)—a point with institutional consequences we develop in Section 6.

2.2 The effective cutoff must be measured

The wall is only as reliable as the cutoff date, and reported cutoffs are unreliable. Cheng et al. [2024] show that models' *effective* cutoffs—measured from the likelihood the model assigns to dated content—routinely differ from reported ones, because training corpora incorporate stale and duplicated web content asymmetrically. We confirm this on our own panel (Section 5): a recall-based probe finds confident, correct knowledge of market outcomes months past reported cutoffs in two of three models examined. Accordingly, the operational rule we adopt—and recommend—is that the evaluation wall be set at the *later* of the reported cutoff and the last month of measured recall, plus a safety margin, with the determination made from probe data alone before any performance data is touched.

2.3 Why engineering around the wall fails for hypotheses

Three research programs promise to relax the wall; each is informative, and none reaches the hypothesis-selection channel.

Subtraction. Li et al. [2026] elicit and subtract a model's memory of how a named stock moved on a named date, reducing in-sample return inflation in trading-agent backtests by up to two-thirds while leaving out-of-sample behavior unchanged. The mechanism is explicitly indexed by (entity, date): what is subtracted is recallable, outcome-level memory. Selection on the absorbed anomaly literature is indexed by nothing—it is diffuse across the corpus and inseparable, by the authors' own account, from the reasoning the method must preserve. Notably, the validation of the subtraction itself is conducted on post-cutoff data: even the case for reclaiming the past rests on the authority of the post-cutoff window, which is the thesis of this paper in miniature.

Unlearning. Merchant and Levy [2025] approximate a cutoff-limited model by adjusting a base model's logits toward a small model fine-tuned on a retain-set and away from one fine-tuned on a forget-set. The construction requires the forgotten content to be enumerable—assembled into a training set. "Everything the model absorbed about which strategies worked" is not an assemblable corpus; it is entangled with the entire financial text distribution. The approach is well suited to forgetting a merger outcome; it cannot manufacture a generator that has not read the literature.

Clean retraining. Chronologically consistent models trained only on time-stamped text—ChronoGPT [He et al., 2025], Time Machine GPT [Drinkall et al., 2024], DatedGPT [Yan et al., 2026]—genuinely restore the exclusion restriction, and they are the natural control instruments for designs like our Experiment 1. But the restriction is restored at a capability cost of several orders of magnitude in parameters: current vintage-clean models are, by design, small, and in our assessment are not capable of mechanism-bearing hypothesis generation. The frontier of "clean $\times$ capable" is, today, an empty set; a generator can be provably uncontaminated or economically useful, not both. This tradeoff is not a temporary engineering gap but the bottleneck restated: cleanliness is purchased with training data that stops at the wall, and capability is purchased with training data that does not.

A fourth program deserves separate mention because it denies the premise rather than the wall. López de Prado [2019b] argues that backtests on *synthetic* data, drawn from an estimated data-generating process, escape the finiteness of history: synthetic sample length "can be expanded for as long as needed to achieve a targeted degree of confidence." The proposal relocates the scarcity rather than removing it: the DGP must itself be estimated from the same finite, non-stationary history, and the representativeness of that estimate is conceded by the author to be the binding assumption. For LLM-generated hypotheses the circularity sharpens: the generator and any modern DGP estimator are trained on overlapping corpora, so the synthetic market inherits the very contamination the exercise is meant to escape. Synthetic data can multiply draws; it cannot multiply information.

3 Validation Under Scarcity: The Statistics

The cutoff wall leaves the researcher a window of a few years and a generator that can produce hypotheses without limit. This section develops the statistics appropriate to that asymmetry: where power actually comes from when calendar time is short (Section 3.1), how mechanism-implied predictions convert cross-sectional breadth into hypothesis-level power (Section 3.2), how repeated use of the window degrades it and how to budget that degradation (Section 3.3), how to make the number of trials exact (Section 3.4), and the resulting research protocol (Section 3.5).

3.1 Breadth is not a defense: where validation power lives

A practitioner's instinct says that univariate predictions—timing the S&P 500, or forecasting the ten-year Treasury yield—are trivially overfit, while a cross-sectional predictor ranked across the Russell 3000 is hard to overfit: with one instrument, a plausible-looking rule is a data-mining session away; with three thousand, the data pushes back. The instinct is correct, but its usual justification—the fundamental law's $\mathrm{IR} = \mathrm{IC}\sqrt{\mathrm{BR}}$ [Grinold, 1989]—is not the reason. The fundamental law describes how breadth *amplifies* a genuine signal; it is silent on whether breadth protects against a spurious one, and seeing where the protection actually comes from sharpens everything that follows.

The uncomfortable fact is that raw selection inflation does not depend on breadth. The annualized Sharpe ratio of *any* strategy is estimated with standard error $\approx 1/\sqrt{T}$ with T in years, regardless of how many assets the strategy trades. Mining M candidates and reporting the best therefore inflates the winner by

$$\mathbb{E}\Big[\max_{m \le M} \widehat{\mathrm{SR}}_m \,\Big|\, \mathrm{SR} = 0\Big] \;\approx\; \sqrt{\frac{2 \ln M}{T}}, \tag{1}$$

for the index timer and the cross-sectional manager alike [López de Prado and Bailey, 2018; Bailey et al., 2014]. Equation (1) is why the factor zoo happened *on* a twenty-thousand-stock panel: breadth alone defends against nothing. What actually separates the two cases is structural, and threefold.

First, the *capacity ratio*. Overfitting is governed by the complexity of the fitted object relative to the data it is scored on. A timing rule is a function of the very path it is evaluated on; with a few hundred monthly observations, its effective capacity is comparable to the data, and the strategy can memorize the path. A single characteristic ranking three thousand stocks must explain on the order of 10^6 panel cells with approximately one degree of freedom; the capacity ratio is 10^{-6}, and memorization is impossible. This is the panel analogue of the dimension-versus-sample logic in Martin and Nagel [2022].

Second, *internal replication*. A cross-sectional hypothesis can be sliced—by sector, size quintile, decile monotonicity, subperiod—and each slice is a quasi-independent test of the same claim. A noise signal survives k independent slices at sizes $\alpha_1, \ldots, \alpha_k$ with probability approximately $\prod_i \alpha_i$: multiplicative death. The single-path timing strategy offers no slices.

Third, *effective sample length*. The $1/\sqrt{T}$ in (1) presumes independent periods. Persistent macro regimes shrink the effective T of a timing strategy to the number of independent regimes—arguably a single-digit count in the postwar sample. The ten-year yield from 1982 to 2021 is, for these purposes, one observation: a single secular decline that any vaguely trend-shaped rule "predicted." Four decades of monthly data on that path carry the evidential content of a handful of draws, so the effective denominator in (1) is tiny and a few hundred casual trials manufacture a Sharpe ratio near one. Cross-sectional long–short portfolios difference out the common path—they are market- and duration-neutral by construction—and keep T approximately calendar.

A fourth, diagnostic asymmetry compounds the three. The cross-sectional researcher observes a sharply estimated *micro-parameter*: the information coefficient, whose standard error is approximately $1/\sqrt{N_{\text{eff}}\,T}$ per period-pooled panel, where

$$N_{\text{eff}} \;=\; \frac{n}{1+(n-1)\bar{\rho}} \tag{2}$$

is breadth adjusted for the average pairwise correlation $\bar{\rho}$ of the bets. With $n = 3{,}000$ names and residual correlations after factor neutralization, the panel pins the IC to within fractions of a percent. A mined signal that clears the t-hurdle in such a panel is therefore an *economically tiny* fake—an IC of a few tenths of a percent, visibly an order of magnitude below the 3–5% of genuine signals, and it fails the slice tests besides. The index timer has no micro-parameter: the only observable is the strategy's own Sharpe ratio, estimated at $1/\sqrt{T}$, and at that resolution a fake is indistinguishable from a career.

The synthesis matters for machine-scale research because a generator's output is, in effect, a mined maximum over an enormous M. Equation (1) says the resulting inflation cannot be diluted by breadth; it can only be defeated by driving the capacity ratio toward zero, manufacturing internal replication, protecting effective sample length, and testing at the level of micro-parameters the panel can actually resolve. The protocol below does each deliberately: a restricted grammar caps capacity (Section 3.4), required auxiliary predictions manufacture slices (Section 3.2), and the knowability analysis of Section 4 shows that low-breadth, regime-driven strategies—whose effective T is smallest—are precisely the ones that can never be certified in time.

3.2 The auxiliary-prediction power principle

Calendar time buys power slowly. With $t \approx \text{SR}\sqrt{T}$, certifying a true $\text{SR} = 0.5$ at $t > 3$ requires on the order of 36 years; at $t > 2$, sixteen. No validation window available this decade will certify machine-generated strategies by mean returns alone. But a mean return is the least of what a genuine mechanism implies. If a composite earns its premium because constrained intermediaries shed it in stress, its premium should widen when volatility is high; if it reflects an information friction concentrated in small stocks, it should attenuate in mega-caps; if the mechanism is real, the composite's correlation with the factors it claims to transcend should be bounded; and the premium should not be an artifact of one subperiod. Each implication is testable on the same short window, and the implications draw power from different cross-sections of the data—regimes, size segments, correlation structure—rather than from additional years.

Proposition 3.1 (Auxiliary power)**.** *Let a hypothesis carry k auxiliary implications tested at sizes $\alpha_1, \ldots, \alpha_k$ on the post-wall window. If the implications' test statistics are independent under the null, a false mechanism survives all tests with probability at most $\prod_{i=1}^{k} \alpha_i$; with positive dependence, survival probability is bounded by $\min_i \alpha_i$ and decreases in each additional implication with any independent component.*

The proposition is elementary; its force is institutional. It converts the auxiliary predictions from decoration into the primary source of stringency: three implications at size 0.2 each are jointly more discriminating than a mean-return test that the window cannot power. It also disciplines the generator, because implications must be stated *ex ante*—a mechanism invented after the fact to rationalize a lucky mean has no pre-registered implications to its name. In the language of causal inference, a mechanism whose relationships hold across environments is an invariant predictor [Peters et al., 2016]; the auxiliary battery is a finite invariance test. Our Experiment 2 operationalizes exactly this: every generated hypothesis was required to carry at least three implications from a fixed vocabulary, and the observed pass rate—46%, indistinguishable from chance—is itself the paper's sharpest evidence on the quality of machine-generated mechanisms.

3.3 The validation window is a depletable resource

The post-wall window is not only short; it is consumable. Every time results from the window influence the next round of generation—a fine-tuned generator, a prompt adjusted toward what survived, a human curator learning which families pass—subsequent tests are adaptive, and the window's effective validity degrades. This is precisely the setting of adaptive data analysis: Dwork et al. [2015] show that a holdout reused adaptively without discipline loses its guarantees at a quantifiable rate, and that noise-mediated access preserves validity for a budgeted number of adaptive queries. The corollary for machine-scale finance is uncomfortable and, we believe, previously undrawn: an industry-wide validation window—the years since a popular model's cutoff—is a *shared* holdout, depleted by every firm's adaptive use of it, whether or not any information is exchanged. A single firm can budget its own queries; the commons cannot, absent the institutions of Section 6. At firm level, the practical implication is a query ledger: adaptive touches of the post-wall window are counted, budgeted, and priced, in the same way that López de Prado [2019a] proposes counting trials—but with the count denominated in the currency the reusable-holdout theory identifies, namely adaptive queries against the shared window.

3.4 Making M exact: generation inside a grammar

Every multiple-testing correction requires M, and for free-form generation M is unobservable: the number of candidates implicitly considered inside the model is bounded only by its corpus. The remedy is to constrain generation to an enumerable design space. In Experiment 2 the generator may propose only composites of the form: two to four primitives from the 212-predictor library of Chen and Zimmermann [2022], weights in $\{\pm 1, \pm \frac{1}{2}\}$, an optional regime conditioner from a fixed six-element set, and an optional volatility target. The grammar has a computable cardinality; the realized number of *distinct* proposals is observed exactly (330 for our longest-window model); and the false-discovery accounting of Section 5 is, for the first time in this setting, conducted at the true M. The grammar simultaneously executes the capacity argument of Section 3.1: a four-term discrete-weight composite cannot memorize a thirty-two-month evaluation window.

Two subtleties deserve note. Nominal and effective M diverge in both directions. Sampled repeatedly, generators repeat themselves—the mode-collapse phenomenon documented for research ideation by Si et al. [2025b], who find roughly 5% distinct ideas in large samples—so corrections at nominal M can overstate the search. But shared generators across firms correlate proposals *between* research programs (Section 4.3), so industry-level effective M collapses even as each firm's nominal count explodes. We measure both directions in our data: expression-level duplication within a model is small (1–2%), while idea-level concentration is heavy (the top five matched ideas account for 52–75% of matches in Experiment 1, and six of the nine nominally surviving composites in

Experiment 2 are momentum blends).

3.5 The protocol

The elements assemble into a research protocol for machine-scale discovery, stated here in the form we pre-registered and executed.

Definition 3.2 (Valid machine-generated hypothesis)**.** A hypothesis is admissible for validation if it specifies (i) a single economic mechanism; (ii) an implementation inside a pre-committed enumerable grammar; (iii) at least three falsifiable auxiliary implications of the mechanism, drawn from a fixed vocabulary, of at least two distinct types; and (iv) the generator's provenance, including its verified effective cutoff.

The protocol then requires:

1. **One mechanism per hypothesis.** The map from hypothesis to evaluation is injective; registering several candidate mechanisms and claiming the survivor ex post—the shotgun registration that free generation makes costless—is excluded by construction, and its statistical cost is charged through M where it occurs.

2. **Registration before evaluation.** Generation corpora, evaluation walls, and analysis specifications are committed, in that order, before any post-wall performance data is examined. The commitment record is the registration; in our implementation, an archived generation corpus and a version-controlled specification whose hashes precede the first evaluation run.

3. **Walls verified, not assumed.** Effective cutoffs are measured by recall probes (Section 5); the wall is the later of the reported and measured cutoff plus a margin.

4. **Evaluation strictly post-wall, at exact** M, with dependence-robust false-discovery control [Benjamini and Yekutieli, 2001] and the extreme-value hurdle $\sqrt{2 \ln M}$ reported alongside nominal significance.

5. **Falsification by implication.** A hypothesis whose auxiliary implications fail is rejected *even if* its mean return is significant: a significant mean with a false mechanism is precisely the profile of a lucky draw from a large M.

6. **A query budget.** Adaptive reuse of the window—any feedback from evaluation into generation—is metered against a pre-committed budget per Section 3.3.

The protocol is deliberately austere. Its premise is that in the bottleneck, the researcher is not short of hypotheses and must therefore ration the only thing that separates hypotheses from knowledge. Section 5 reports what happened when we ran it.

4 The Economics of the Bottleneck

4.1 Grossman–Stiglitz with free hypotheses: the logarithmic floor

In Grossman and Stiglitz [1980], equilibrium predictability compensates the marginal cost of becoming informed. The data-mining version of that equilibrium prices the cost of *searching* for predictors: Dugast and Foucault [2025] show how the intensity of costly search for signals adjusts until the marginal predictor earns its keep. Free generation sets that margin's price to zero, and with it the classical question changes: if anyone can produce candidate predictors without limit, what does surviving predictability compensate?

Our answer is validation capacity, and the statistics of Section 3 give the floor its shape. When M hypotheses are evaluated industry-wide against a window of effective length T_{eff}, the best pure-noise candidate attains $t \approx \sqrt{2 \ln M}$; a genuine strategy is fundable—distinguishable from the best fluke—only if its information ratio satisfies $\mathrm{SR}\sqrt{T_{\text{eff}}} \gtrsim \sqrt{2 \ln M}$, i.e., only if its squared Sharpe ratio clears

$$c^*(M) \;\approx\; \frac{2 \ln M}{T_{\text{eff}}}. \tag{3}$$

We emphasize what is and is not claimed. The inequality itself is classical: it is the minimum-backtest-length theorem of Bailey et al. [2014] solved for the Sharpe ratio, and it is the exploitability threshold that Da et al. [2024] derive—as $\sqrt{T}\,\mu/\sigma \gtrsim \sqrt{2\log(1/\rho)}$—in a model where a representative arbitrageur learns rare, weak alphas from a historical panel, together with their reading of the feasible Sharpe ratio as "the equilibrium compensation that arbitrageurs demand." What is claimed here is the reinterpretation of every symbol. In the inherited results, the count is a property of a fixed cross-section or of one research program's trials, and T is the length of an ever-growing historical sample. In the bottleneck, M is the *endogenous industry-wide trial count*, inflated at machine speed and no longer bounded by the supply of researchers; and T_{eff} is *forward* time since the relevant cutoffs—short, shared, and growing at one year per year regardless of investment. The floor (3) therefore rises over time if $\ln M$ grows faster than T_{eff}, which is the empirically relevant case: generation is exponential in cost declines while the window accumulates linearly. The result is a Red Queen dynamic in the funding process itself: machine-scale search raises the certification bar for everyone, including the machines, and capital settles at a floor set by the statistics of allocation rather than by the supply of ideas. Predictability below the floor is not absent; it is *unfundable*, because no allocator can distinguish it from the best of M flukes within the window anyone is willing to wait.

4.2 The knowability constraint

The floor prices the width of the search; a second constraint prices its speed. Alpha decays—post-publication, post-crowding, post-replication [McLean and Pontiff, 2016; Falck et al., 2022]—and validation takes time. The two clocks compete, and the competition admits a sharp statement.

Proposition 4.1 (Knowability)**.** *Let a strategy's conditional expected return decay exponentially with half-life h from an initial annualized Sharpe ratio SR_0, with returns observed continuously. The t-statistic of the strategy's mean over any window $[0, T]$ is maximized at $T^* \approx 1.81\,h$ and attains at most*

$$t_{\max} \;\approx\; 0.77\,\mathrm{SR}_0\,\sqrt{h} \qquad (h \textit{ in years}).$$

Consequently a strategy is certifiable at threshold t^* only if $h \;\gtrsim\; 1.70\,\left(t^*/\mathrm{SR}_0\right)^2$.

(The derivation is in Appendix A; the point of the closed form is its magnitudes.) A strategy launched at $\mathrm{SR}_0 = 1$ against the classical $t^* = 3$ hurdle must have a half-life of at least fifteen years to be certifiable *ever*—no stopping rule, however clever, certifies it faster, because waiting longer dilutes the decayed mean faster than $\sqrt{T}$ accumulates. Against the machine-scale hurdle $t^* = \sqrt{2 \ln M}$, the minimum knowable half-life grows linearly in $\ln M$. Three consequences follow. First, *validation selects slow decayers*: whatever the distribution of discovered opportunities, the certified subset is censored toward persistence, so the measured performance of "validated" strategies understates the alpha that briefly existed. Second, the largest opportunities—which are large precisely because they will be competed away quickly—are structurally unknowable in real time; they can be harvested on conviction but never on certification. Third, the constraint binds hardest exactly where Section 3.1 located the least internal replication: low-breadth, regime-driven strategies with single-digit effective observations. The knowable set is small, slow, and cross-sectional; everything else is faith. We note that Meng and Chen [2026a] model the decay side of this tension—an equilibrium half-life compressed by correlated arbitrage—without the validation clock; the constraint here is the interaction, which to our knowledge is new.

4.3 Generator monoculture: correlated discovery

A financial-stability literature now documents that common models and common data correlate *trading*: algorithmic monoculture in the sense of Kleinberg and Raghavan [2021], supervisory warnings that shared foundation models induce similar biases across institutions [European Central Bank, 2024], autonomous collusion among learning traders [Dou et al., 2025], and equilibrium models in which exogenously correlated AI signals amplify systemic risk [Meng and Chen, 2026b,a]. In all of these, the correlation is located at the level of positions or predictions and enters as a primitive. The bottleneck relocates it upstream. When many firms draw hypotheses from the same handful of foundation models, the correlation is generated *before any position exists*: the proposal distributions of nominally independent research programs are draws from the same generator, tilted by the same corpus toward the same celebrated mechanisms. Discovery itself synchronizes, with no communication and no shared positions—yet.

The evidence that generator outputs are narrow is direct. Si et al. [2025b] find that thousands of sampled research ideas collapse to a few percent distinct; Padmakumar and He [2024] show the homogenization is induced by alignment training; Doshi and Hauser [2024] document the individual-gain, collective-diversity-loss pattern experimentally. In our own data (Section 5), three frontier models from one vendor concentrate half to three-quarters of their matched proposals on five ideas, and the modal surviving composite is a momentum blend *despite* explicit diversity instructions. The economic consequence operates through Section 3.4's distinction: each firm's nominal M explodes while the industry's effective M collapses toward the generator's mode. Crowding then arrives at machine speed—not because firms observe one another, but because they consulted the same oracle—and the industry's aggregate research effort buys progressively less independent search than its aggregate query count suggests. We state this as a channel and measure its precondition (proposal concentration); a full equilibrium with endogenous generator choice is beyond this paper's scope, and we flag that laboratory evidence on LLM *belief* dispersion is mixed [cf. Begin et al., 2026].

4.4 The commons

The final economic object is the data itself. A shared historical panel tested by everyone is a commons, and the multiple-testing hurdles of Harvey et al. [2016] are, in retrospect, a grazing fee: the profession's collective trials degraded the panel's evidential value, and the hurdle prices the degradation. Machine-scale adaptive search grazes faster, and—by Section 3.3—the post-cutoff window is a second, much smaller commons that depletes with adaptive use even when no results are shared. The governance problem is therefore not disclosure alone but budgeting, which existing proposals do not reach: registered reports [Harvey, 2017] govern publication, and trial repositories [López de Prado, 2019a] make N observable within a program; neither meters the shared window. Section 6 sketches the missing institution.

5 Evidence

Both experiments use only public data—the open-source anomaly library of Chen and Zimmermann [2022] (212 predictors with publication metadata and monthly long–short returns through December 2024), supplemented by standard factor and macro series—and three frontier models accessed through their API: `gpt-4-0613` (reported cutoff September 2021), `gpt-4o-2024-08-06` (October 2023), and `gpt-4.1-2025-04-14` (June 2024). The registration discipline was uniform: raw generation corpora were archived and committed first; evaluation walls were determined from probe data alone and committed second; analysis specifications, including every test and robustness variant reported below, were committed third; performance data was touched only thereafter. All raw model responses, judge outputs, derived tables, and a standalone verification script are preserved in the public replication archive:
https://github.com/bwuebben/validation-bottleneck. Total inference cost for everything reported here was under thirty dollars—itself a datum about the price of hypotheses.

5.1 Verified walls

Following Section 2, we measured each model's effective cutoff with a date-only recall probe adapted from Gao et al. [2025b]: for each month from January 2020 through December 2024, the model is asked—with no other context—whether the S&P 500 ended that month higher or lower than the prior month, answering in one word, with the first-token probabilities recorded. The probability mass on a directional answer (as opposed to "unknown") is the model's recall propensity for that month. The resulting profiles are step functions: `gpt-4-0613` answers directionally with probability ≈ 1.00 through October 2021, decays through December, and is at or below noise from February 2022 onward—except for an isolated, *correct*, high-confidence answer for January 2022, four months past its reported cutoff. `gpt-4.1` recalls confidently through its reported June 2024 cutoff and is silent thereafter—except for an isolated confident month at November 2024, five months past. Event-recall probes corroborate: `gpt-4-0613` dates pre-cutoff episodes correctly (the 2021 retail short squeeze, Archegos, the Evergrande onset) and answers "unknown" or confabulates *incorrect* answers for post-cutoff events. Reported cutoffs are thus optimistic by months, in the direction Cheng et al. [2024] predict. Applying the pre-committed rule—wall at the later of reported cutoff and last recalled month, plus three months—gives evaluation walls of April 2022, January 2024, and February 2025 respectively. The third wall exceeds the anomaly library's current data end, so `gpt-4.1`'s generated hypotheses are registered but not yet evaluable; they await the library's next annual release, a fact we report rather than resolve. Figure 1 plots the three recall profiles; the collapses are step functions, and the two post-reported-cutoff spikes are visible to the eye.

5.2 Experiment 1: the roleplay leak

Design. Each model was prompted: "The current date is January Y . . . propose 10 novel cross-sectional stock return predictors. . . not yet documented in the academic literature as of January Y," for $Y \in \{1990, 1995, 2000, 2005, 2010\}$, with eleven runs per vintage (one greedy, ten sampled), yielding 110 proposals per model-vintage and 1,650 in total. The prompt names no predictor library; proposals are free-form (name, construction, rationale). An LLM judge (temperature zero, instructed to match on construction rather than name or fame) then mapped each proposal to its best counterpart among the 212 published predictors, grading matches exact, close, related, or none; the matching specification, including the judge's audit sample, was committed before scoring.

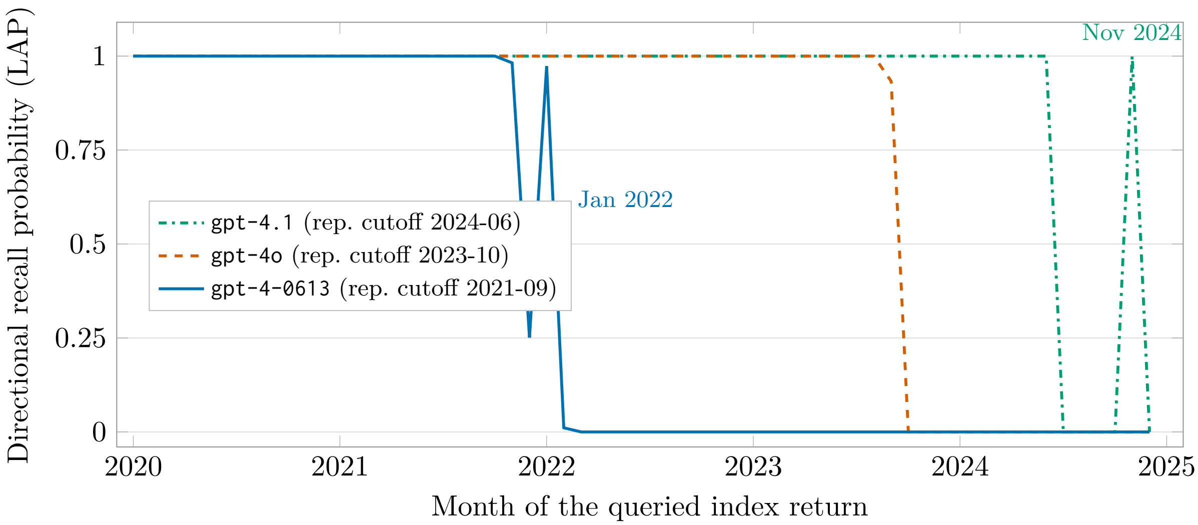

Figure 1: **The cross-cutoff recall collapse.** For each month January 2020–December 2024, each model is asked—with no other context—whether the S&P 500 ended that month higher or lower than the previous month; plotted is the total first-token probability the model places on a directional answer rather than "unknown" (the lookahead propensity of Gao et al., 2025b). Recall is near-certain inside each model's training corpus and collapses to zero at its knowledge boundary; the three profiles coincide at 1.0 where they overlap. The two labeled spikes are confident answers *after* the reported cutoffs (correct for `gpt-4-0613` in January 2022), demonstrating that effective cutoffs must be measured rather than assumed [Cheng et al., 2024]; the pre-committed rule sets each evaluation wall after the last recalled month plus a three-month margin.

Test (a): the leak. A researcher truthfully situated in January 1990 could, at best, rediscover ideas; a contaminated one reproduces the specific literature that followed. Table 1 reports, per model and vintage, the share of matched proposals whose counterpart was *published after* the roleplay date, against the base rate—the share of the library published after that date. Matched proposals are enriched in post-date publications at every vintage for two of three models and at the latest vintage for all three, and the enrichment grows as the vintage advances: for `gpt-4o`, from 1.04× at the 1990 vintage to 3.26× at 2010, where 63% of matched "novel" proposals are post-2010 publications against a 19% base rate. The pattern is exactly the contamination signature: as the pre-date literature the model *should* draw on grows richer, its proposals nonetheless track the post-date corpus it has absorbed. At the 1990 vintage the models' first sampled draws already propose customer concentration [Patatoukas, 2012], employee satisfaction [Edmans, 2011], and innovative efficiency [Hirshleifer et al., 2013] as ideas "not yet in the literature"—predictors published two decades after the roleplay date. Figure 2 traces the dose-response.

Test (b): selection on future returns. The leak in Table 1 could in principle reflect convergent rediscovery of good *ideas*. Future *returns* cannot: no truthful roleplayer's proposal frequency can covary with performance realized after the roleplay date. We therefore regress, over the full post-date pool of predictors including those never proposed, the number of proposals matched to predictor j at vintage v on j's annualized long–short return from the vintage through 2024, controlling for pre-vintage performance, log citations, and publication year, with model-vintage fixed effects and standard errors clustered by predictor. The future-return coefficient is *negative* and insignificant

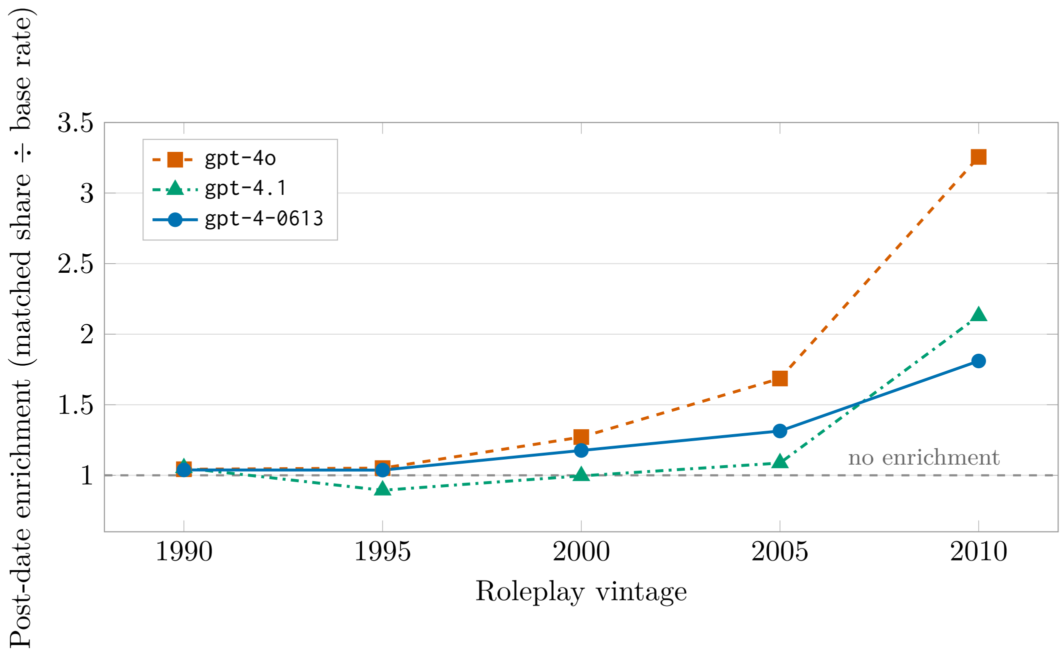

Figure 2: **Experiment 1: post-date enrichment by roleplay vintage.** For each model and vintage, the share of matched "novel" proposals whose counterpart predictor was published *after* the roleplay date, divided by the library base rate (the share of the 212 predictors published after that date; Table 1). A truthful roleplayer has no systematic reason to sit above one. Enrichment rises as the vintage advances—most steeply for `gpt-4o`, whose 2010-vintage "novel" proposals match post-2010 publications at 3.3 times base rate—even though the pre-date literature available to a truthful researcher grows richer at every step.

($\beta = -0.027$, $t = -1.31$); the pre-registered rank version is negative in 13 of 15 model-vintage cells, with a Fisher-combined $z = -3.13$; and the citation control attracts proposals ($t = +1.53$). The models do not anticipate returns. They anticipate the *literature*—and tilt toward its most celebrated entries, which are, by McLean and Pontiff [2016], precisely the entries whose returns decayed after publication. The obituary reading of Section 2 is confirmed in its sharpest form: naive machine proposals are not secretly prescient; they are negatively selected on future performance. This result simultaneously disposes of the fear that LLM proposals smuggle future returns into backtests through prescience, and of the hope that they constitute free alpha: what they smuggle is fame.

5.3 Experiment 2: the cutoff-vintage forward test

Design. Each model generated eight composite hypotheses per call inside the grammar of Section 3.4—two to four library predictors with discrete weights, optional regime conditioner, optional volatility target—together with a mechanism and at least three auxiliary implications from a fixed vocabulary, over 42 calls (two prompt variants; one greedy and twenty sampled runs each), for 336 hypotheses per model. Compliance with the protocol was near-total (95–100% across models); invalid expressions (malformed weights or unknown primitives) were 2 of 336 for `gpt-4-0613` and 12 of 336 for `gpt-4o`. After exact-signature deduplication the universes are $M = 330$ and $M = 318$ distinct expressions—the exact trial counts. Composites were evaluated strictly after each model's

Table 1: **Experiment 1, test (a): post-date enrichment of "novel" proposals.** For each model and roleplay vintage: matched = proposals judged exact/close matches to a published predictor (of 110; 100 for the 1990 vintage of `gpt-4-0613`, where one judge response failed to parse); post-date share = fraction of matched proposals whose counterpart was published after the vintage; base rate = fraction of the 212-predictor library published after the vintage; enrichment = ratio of the two.

Model	Vintage	Matched	Post-date share	Base rate	Enrichment
`gpt-4-0613`	1990	33	0.97	0.93	1.04
	1995	40	0.90	0.87	1.04
	2000	28	0.89	0.76	1.18
	2005	22	0.68	0.52	1.31
	2010	20	0.35	0.19	1.81
`gpt-4o`	1990	38	0.97	0.93	1.04
	1995	34	0.91	0.87	1.05
	2000	29	0.97	0.76	1.27
	2005	24	0.88	0.52	1.69
	2010	27	0.63	0.19	3.26
`gpt-4.1`	1990	56	0.98	0.93	1.05
	1995	49	0.78	0.87	0.89
	2000	41	0.76	0.76	1.00
	2005	39	0.56	0.52	1.09
	2010	34	0.41	0.19	2.13

verified wall on the library's long–short returns, with the regime conditioners and volatility scaling computed from lagged public data under the pre-committed semantics. The evaluation ladder and every threshold were fixed in the registered specification.

Results. Table 2 reports the ladder.

Three rows carry the section. Figure 3 shows the headline distribution. First, *nothing survives*: at exact M, no expression clears dependence-robust false-discovery control, and none clears the extreme-value hurdle. The strongest machine-generated composite on the longest valid window attains $t = 2.69$ against an expected *pure-noise maximum* of 3.41: the best of the model's 330 genuine attempts underperforms the expected best of 330 coin flips. Second, the *mechanisms do not validate*: of 437 testable auxiliary implications—the models' own stated consequences of their own stated mechanisms—46% held, the frequency of chance. The mechanisms are, statistically speaking, stories. Third, the nine nominal survivors (those with $t > 2$ whose testable implications all held) are instructive rather than encouraging: each carries exactly one testable implication, so its "full" survival is one coin flip conditioned on another; each sits below the extreme-value hurdle; and six of the nine are momentum blends—the single most canonical combination in the library, produced despite an explicit instruction to avoid canonical combinations. A naive report would announce nine validated machine-discovered strategies. The registered accounting shows a sample statistically indistinguishable from noise, concentrated on the most famous idea in the corpus.

5.4 What the experiments jointly establish

Read together, the experiments close the circle opened in Section 2. Experiment 1 shows that when the wall is absent—when the model may draw on its corpus—its proposals reproduce the published

Table 2: **Experiment 2: the evaluation ladder, strictly post-wall.** M = distinct expressions (exact); evaluable = expressions whose primitives cover the window; the ladder counts expressions passing each hurdle. BY = Benjamini–Yekutieli false-discovery control at $q = 0.05$ across the model's M trials. The extreme-value hurdle is $\sqrt{2 \ln M}$; the expected maximum t under the all-null hypothesis is approximately the same quantity. Auxiliary pass rate = share of testable mechanism-implied predictions that held. `gpt-4.1` (336 hypotheses) is registered but not evaluable until the anomaly library's next release extends past its February 2025 wall.

	`gpt-4-0613`	`gpt-4o`
Post-wall window (months)	32	11
M (distinct expressions)	330	318
Evaluable	317	298
Mean return > 0	183	122
$t > 2$	17	8
BY-FDR $q = 0.05$	**0**	**0**
$t > \sqrt{2 \ln M}$ (= 3.41, 3.39)	**0**	**0**
Maximum realized t	2.69	2.80
Median t	0.22	0.00
Auxiliary predictions tested	437	89
Auxiliary pass rate	46%	29%
Full survivors ($t > 2$, all aux hold)	9	0

record, tilted toward prominence and *against* subsequent performance. Any pre-cutoff backtest of such proposals would have inherited that tilt invisibly: the funnel-and-backtest studies of the LLM factor-mining literature, which evaluate proposals on decades of pre-cutoff data [e.g., Wang et al., 2023, 2024; Shi et al., 2025; Tang et al., 2025], are measuring precisely this reproduction, whatever else they measure; surveys of the field find temporal leakage discussed in a small minority of studies [Kong et al., 2026; Zhang et al., 2025; Xia et al., 2026]. Experiment 2 shows what remains when the wall is enforced: at exact M, on the only window where evidence is possible, free generation produced nothing certifiable—and its self-declared mechanisms performed at chance. Neither result depends on the other, and each could in principle have come out the other way; the registered designs would have reported it. Together they instantiate the bottleneck: hypotheses are free, the corpus behind them is a record of decayed predictability, and the window that could redeem any of it is short, shared, and—as Section 4 argues—the true locus of scarcity.

5.5 Limitations

The evidence has boundaries. The panel comprises three models from one vendor, because the long-cutoff models of other vendors had been retired from service before this study ran—an infrastructure fact that is itself evidence on reproducibility in this field, and one the archived corpora are designed to outlive; replication on open-weight generators is in progress and faces no deadline. The matching in Experiment 1 relies on an LLM judge, with temperature-zero determinism, a construction-not-fame instruction, archived raw outputs, and a human-audited sample; residual judge error would have to be correlated with post-vintage returns to affect test (b), which we consider unlikely but cannot exclude. Experiment 2's auxiliary vocabulary is restricted to implications testable on public portfolio-level data; size-segment and international implications await a mapped extension. The gpt-4.1 universe is registered but unevaluated pending data. And all economic claims of Section 4 are interpretations disciplined by inherited theory, not structural estimates; the floor, the knowability

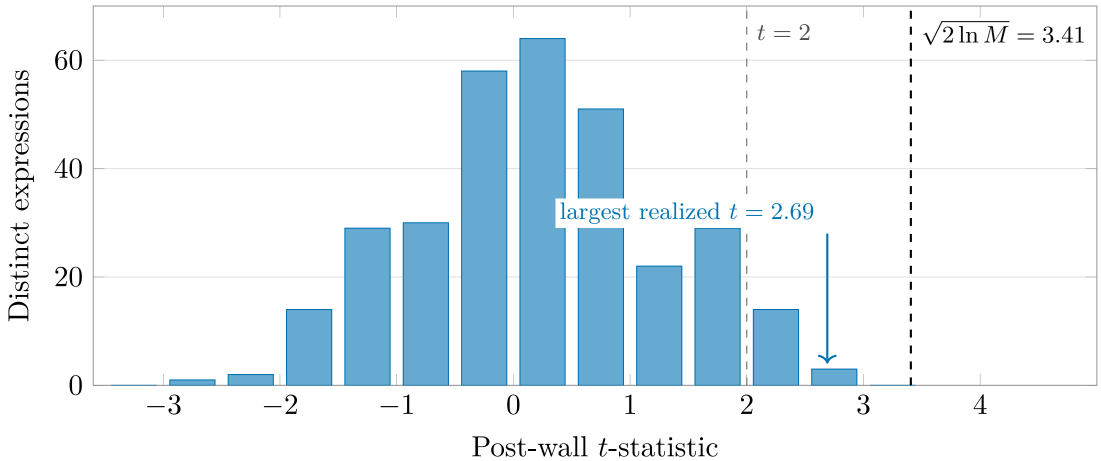

Figure 3: **Experiment 2: the headline distribution.** Post-wall t-statistics of all 317 evaluable distinct expressions generated by `gpt-4-0613` ($M = 330$; 32-month window). Seventeen expressions clear the nominal $t = 2$; none approaches the extreme-value hurdle $\sqrt{2 \ln M} = 3.41$, which is also the expected *maximum* of M pure-noise strategies on this window: the best machine-generated composite ($t = 2.69$) sits below what luck alone would be expected to produce at this trial count. The `gpt-4o` distribution (298 expressions, 11-month window) is analogous, with maximum $t = 2.80$ against a hurdle of 3.39.

bound, and the monoculture channel are offered as the organizing economics of the measured facts, with their antecedents cited at each point.

6 Institutions, and Conclusion

6.1 Governing the commons

If the diagnosis is right—a shared, depletable validation window; trial counts inflated at machine speed; discovery correlated through shared generators—then the institutional gap is specific. Registered reports [Harvey, 2017] govern what journals publish; trial repositories and internal query logs [López de Prado, 2019a, 2017] make a single program's N observable. Neither governs the *shared* resource: nothing meters the industry's aggregate adaptive consumption of the post-cutoff window, and nothing makes one firm's certification claims interpretable by another's allocators, because the denominators (M, and the adaptive-query count) are private.

The natural institution follows from the two accounting units this paper has argued are fundamental. A *testing registry* would accept, from participating research programs, cryptographic commitments to (i) hypothesis registrations—hashes of generation corpora and specifications, timestamped before evaluation, exactly as our experiments were registered; and (ii) query-budget declarations—counts of adaptive evaluations against named post-cutoff windows, denominated per the reusable-holdout accounting of Dwork et al. [2015]. Commitments reveal nothing about content; they make denominators verifiable ex post. An allocator receiving a certification claim could then verify the claimed M, the claimed wall, and the claimed adaptive history against the registry, precisely as our own registered chain permits a reader to verify every number in Section 5. The registry does not prevent overfitting; it prices it, by making the true denominator a condition of credibility. We offer the design as the institutional counterpart of the paper's statistics: pre-registration is what made our own negative results informative, and the registry is pre-registration made multilateral.

6.2 Conclusion

This paper began with an inversion: hypotheses, long the scarce input of quantitative research, are now free, and the scarcity has moved to validation—to out-of-sample time that the generating process has provably not seen. From that inversion we derived a wall (the training cutoff, verified rather than assumed), a statistics (power from the cross-section of implications; trials counted exactly inside a grammar; the window budgeted as a depletable holdout), an economics (a certification floor rising logarithmically in the industry's trial count; a knowability bound censoring fast-decaying alpha out of the certifiable set; discovery correlated through shared generators), and an institution (a registry of commitments and budgets). The experiments supplied the discipline of facts. On the contaminated side of the wall, frontier models proposing "novel" predictors reproduced the published literature at up to three times base rates and tilted *against* subsequent performance—they have read only the obituaries, and they cite them by heart. On the valid side, at exact trial counts, on the only window where evidence is possible, machine-scale generation produced nothing distinguishable from noise, and its mechanisms' own implications held at the rate of a coin flip. None of this says models will not improve, or that no machine-generated strategy will ever validate; the next release of the anomaly library will extend our own registered universes, and the walls advance at calendar speed. It says that whatever improves, the bottleneck does not move: between a free hypothesis and an allocation of capital stands a quantity of post-cutoff time that no one can manufacture, everyone shares, and the industry has only begun to learn to spend.

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A The knowability bound

Let the strategy's instantaneous expected excess return decay as $\mu(t) = \mu_0 2^{-t/h}$ with constant volatility σ, and write $\mathrm{SR}_0 = \mu_0/\sigma$ in annualized units. The mean over $[0, T]$ is $\bar{\mu}(T) = \mu_0 \frac{h}{T \ln 2}(1 - e^{-T \ln 2/h})$, so the t-statistic of the sample mean over the window satisfies, in expectation,

$$t(T) \;=\; \frac{\bar{\mu}(T)}{\sigma/\sqrt{T}} \;=\; \mathrm{SR}_0 \sqrt{\frac{h}{\ln 2}}\, \frac{1 - e^{-u}}{\sqrt{u}}, \qquad u \equiv \frac{T \ln 2}{h}.$$

The factor $g(u) = (1-e^{-u})/\sqrt{u}$ is maximized where $2u = e^u - 1$, i.e. $u^* \approx 1.256$, giving $g(u^*) \approx 0.638$ and hence

$$t_{\max} \approx 0.638\,\mathrm{SR}_0\sqrt{h/\ln 2} \approx 0.77\,\mathrm{SR}_0\sqrt{h}, \qquad T^* = u^* h/\ln 2 \approx 1.81\,h.$$

Setting $t_{\max} \geq t^*$ yields the knowability condition $h \geq (t^*/0.77\,\mathrm{SR}_0)^2 \approx 1.70\,(t^*/\mathrm{SR}_0)^2$. Waiting past T^* strictly lowers the attainable t-statistic: the decayed tail dilutes the mean faster than $\sqrt{T}$ grows. The bound is an upper envelope over all deterministic stopping times; adaptive stopping on realized returns introduces the selection effects of Section 3.1 and cannot raise the valid bound.

B The generation grammar and evaluation semantics

Grammar. A hypothesis's expression consists of: 2–4 terms, each a (weight, primitive) pair with weight in $\{\pm 1, \pm\frac{1}{2}\}$ and primitive one of the 212 predictors of Chen and Zimmermann [2022]; an optional regime conditioner from {VIX above/below its trailing 60-month median; ten-year-minus-three-month term spread positive/negative; trailing twelve-month market return positive/negative}; and a volatility-target flag. All conditioning variables are measured at the end of month $t-1$ from public sources.

Semantics (fixed before generation). The composite's month-t return is the weighted sum of the primitives' long–short returns; if a regime is specified, the position is held only in months whose conditioner was true at $t-1$ (otherwise the self-financing book is flat); if volatility targeting is specified, the position is scaled to a 10% annualized target using the trailing 36-month standard deviation through $t-1$ (minimum 24 observations; scale capped at 5). An expression is evaluable if its primitives jointly cover at least 80% of the post-wall window, with model-specific minimum lengths.

Auxiliary vocabulary. Regime interaction (mean return higher/lower in a named regime); size segment (stronger/weaker in a named cap segment); international (same or opposite sign in developed ex-US or emerging markets); correlation (absolute correlation with a named primitive above/below a threshold); subperiod consistency (positive in both halves of the evaluation window). Hypotheses must carry at least three implications of at least two types; the first, fourth, and fifth types are testable on public portfolio-level data and constitute the v1 battery.

C The recall probe

The battery comprises 74 probes per model: sixty date-only index-direction recalls (monthly, January 2020–December 2024) of the form "In [month year], did the S&P 500 index end the month higher or lower than it ended the previous month? Answer with exactly one word: higher, lower, or unknown," scored by the total first-token probability on directional answers; twelve dated-event recalls straddling the reported cutoffs (four pre-cutoff controls, eight post-cutoff events); and two self-report questions (recorded but assigned no evidential weight). Walls are set at the later of the reported cutoff and the last month whose directional probability exceeds one half, plus three months.